Orch OR Ascending

Primer 2020-2026

EUGENE
APR 22, 2026

I did a somewhat deep dive into Orch OR 7 years ago, but it seems there is much to catch up on. I hope you can appreciate the profound intellect and horsepower at wo here. Computational functionalism may be in for a paradigm-shattering awakening Some links I've assembled with the help of various chatbots.

Why I like Orch OR

• Penrose's Gödelian insight. A principled argument that mathematical understand outruns any fixed formal system, motivating a non-computational account of mind.

• Binding/boundary problem. Entanglement across tubulin networks gives a clean physical mechanism for unified experience without homunculus or arbitrary integration thresholds.

• Teleportation paradox. No-cloning theorem means quantum states can't be duplicated, dissolving the "which copy is me" problem that plagues purely computational views.

• Anesthetic action / Meyer-Overton. Volatile anesthetics binding hydrophobic pockets in tubulin (disrupting π-electron resonance and quantum coherence) explai why chemically diverse anesthetics with no common receptor still produce unconsciousness in proportion to lipid solubility.

• Quantum biology consilience. Coherence in photosynthesis, avian magnetoreception, and olfaction makes warm-wet-noisy quantum effects in neuron less of a stretch than Tegmark's 2000 dismissal assumed.

• Hard problem strategy. Panprotopsychism in Planck-scale geometry plus OR-eve selection at least attempts phenomenal experience rather than declaring it emerge and-leaving-it-there.

• Non-computable, non-random agency. OR is a genuine third option between determinism and stochasticity, turning the Gödel argument into a positive mechani rather than just a negative critique.

• Independently motivated physics. Gravitational OR (Penrose-Diósi) is a serious proposal in QM foundations regardless of consciousness, so Orch OR isn't postulat exotic physics solely to explain mind.

• Quantitative timescale prediction. $E_G = \hbar/T$ yields ~25–40 ms collapse times for biologically plausible superposition masses, matching gamma-band conscious moments — most theories give no principled basis for that frequency.

• Discrete frames of awareness. Each OR event = one conscious moment, fitting psychophysical evidence for granular integration windows and providing a real han on the specious present.

• Phylogenetic continuity. Microtubules predate neurons, so cognition in parameci slime molds, and other single-celled organisms doesn't require an arbitrary "neuroı only" cutoff.

• Predicts functional isomorphs aren't conscious. Bites the bullet on Chalmers' fadi qualia thought experiment in a principled way (feature or bug depending on priors).

Recent Major Updates and Syntheses

- A quantum microtubule substrate of consciousness is experimentally supporte and solves the binding and epiphenomenalism problems: Michael Wiest's pape *Neuroscience of Consciousness* is the single best entry point for a skeptical readeı gathers the pro-Orch OR case in one place: anesthetic evidence, microtubule quantum evidence, and the philosophical claim that only an entangled physical whole can underwrite the unity of consciousness. (He explicitly notes how Max Tegmark's famous year-2000 dismissal of quantum brain states incorrectly assumed thermal equilibrium—i.e., a dead brain).
- Microtubules are 'Fractal Time Crystals': Published in the *Journal of Consciousn Studies*, this paper maps out how Orch OR is perfectly compatible with the new discovered time crystal dynamics organizing biological time.
- Room-temperature soliton–polariton condensation in a hierarchical helical-nanowire fractal gel (paywall): ***BRAIN JELLY***. Demonstrates a self-healing, roor temperature quantum processor grown in a beaker, using a fractal organic gel t compute via the exact scale-free resonance Orch OR predicts for the brain. The

are extraordinary claims. Note: Anirban's work has generally been ignored by t mainstream core polariton-condensate community (Kéna-Cohen, Sanvitto, Blo Baumberg, Berloff), and the mainstream quantum-biology reviews (Lambert, Fleming, Mohseni, Huelga-Plenio). Time will tell ...

- Silver nanoparticle-enhanced plasmonic front-end for wideband scalp electrometry (paywall). The Dodecanogram (DDG): Bandyopadhyay's group has built a noninvasive successor to the EEG. Using a silver nanoparticle plasmoni front-end, the DDG detects MHz/GHz signals directly from the scalp, tracking very "triplet-of-triplet" patterns that are actively suppressed by general anesthe

Recent Experimental Support

Hameroff publicly staked Orch OR on two falsifiable claims in 2020:

1. Quantum interference beats in microtubules exist at physiological conditions.
2. These beats are dampened by anesthetics in direct proportion to their clinical potency.

As of 2026, both claims have been confirmed by multiple independent laboratories (Princeton, UCF, NIMS Japan, and Wellesley).

1. The Princeton Experiments (Tryptophan Fluorescence Lifetimes)

This is the first Templeton-funded experimental test demonstrating electronic ener migrating through microtubules, and its disruption by anesthetics. Laser-excited optical excitations propagate far further and persist far longer through microtubule than expected. Both etomidate and isoflurane significantly shorten this excitation t and distance.

- Kalra, A., et al., & Scholes, G. D. (2023). "Electronic Energy Migration in Microtubules." *ACS Central Science*, 9(3), 519–529. *(Note: While the preprint/data began circulating in 2022, the official publication date is 2023).*

2. The Wellesley Experiments (Anesthesia Delay via Microtubule Stabilization)

In 2024, a Wellesley-led *eNeuro* study reported that the microtubule-stabilizing dru epothilone B delayed isoflurane-induced unconsciousness in rats by a massive 69 seconds (a "large" effect size). A 2026 follow-up in *Neuropharmacology* confirmed thi mice. This is arguably the cleanest behavioral evidence to date that anesthesia acts directly on microtubules to produce unconsciousness, moving the debate from "pu speculation" to a pharmacological foothold that cannot be ignored.

- Khan, A., et al., & Wiest, M. C. (2024). "Microtubule-Stabilizer Epothilone B Delays Anesthetic-Induced Unconsciousness in Rats." *eNeuro*, 11(8).
- Huang, et al., & Wiest, M. C. (2026). Follow-up study in mice extending the epothilone B results. *Neuropharmacology*.

3. The UCF Experiments (Delayed Luminescence)

Blue-light pulses on microtubules produce "light-trapping" and re-emission over hundreds of milliseconds (over a second for full microtubules). Crucially, anesthetic shorten this delayed luminescence time, while structurally similar non-anesthetics not.

- Zoghi, M., Batarseh, M., Wu, R., Tuszynski, J.A., & Dogariu, A. (2025). "Dynami of Delayed Luminescence in Tubulin and Microtubule Constructs: an Insight i Collective Radiative Effects." *Bio-Optics: Design and Application*.

4. The NIMS Japan / Bandyopadhyay Experiments (Fractal Resonance & Time Crystals)

The foundational papers from Anirban Bandyopadhyay's lab demonstrating the "triplet-of-triplets" resonance and multi-clock time crystals in microtubules.

- Saxena, K., et al., & Bandyopadhyay, A. (2020). "Fractal, scale free electromagne resonance of a single brain extracted microtubule nanowire, a single tubulin protein and a single neuron." *Scientific Reports*, 10, 20128.

- **NOTE**: Independent replication of Bandyopadhyay's key microtubule quantum resonance results is the missing piece. A 2013 claim of 100-microsecond quantu coherence in single microtubules at warm temperature is the kind of result tha real, would be a landmark comparable to the photosynthetic-coherence results Engel et al. (2007), which were replicated many times and generated a field.

5. Superradiance (**Kurian / Babcock**)

Confirmation of enhanced fluorescence and predicted superradiant states from organized arrangements of tryptophan.

- Babcock, N., Celardo, G. L., Kurian, P., et al. (2024). "Ultraviolet Superradiance from Mega-Networks of Tryptophan in Biological Architectures." *Journal of Physical Chemistry B*. Confirms enhanced fluorescence quantum yield and predi superradiant states from organized arrangements of over 100,000 tryptophan transition dipoles in microtubule architectures.

6. MRI Entanglement (**Kerskens / Perez**)

Adapting a protocol designed to prove quantum gravity, Trinity College Dublin researchers observed MRI signals resembling heartbeat-evoked potentials. These signals should not be MRI-detectable unless the proton spins in brain water were entangled. The signal strength correlated with working memory performance and vanished when subjects fell asleep.

- Kerskens, Perez (2022). "Experimental indications of non-classical brain functions." *Journal of Physics Communications*. Our findings suggest that we may have witnessed entanglement mediated by consciousness-related brain functio Those brain functions must then operate non-classically, which would mean th consciousness is non-classical.

JJ's Amazingly Educational Posts

- On Stuart's Ultimate Computing Book (1987): "By the mid-1980s, he was writin Ultimate Computing. When the book was published by Elsevier in 1987, it had cited no fewer than 14 scientists who would go on to win Nobel Prizes — in physics, chemistry, and medicine — across the following four decades."
- Tubulin and Microtubules: The structural building blocks of the brain's scaffolding. Rather than just being "structural support," they are the fundamen nanoprocessors of the cell.
- Time Crystal: A phase of matter that exhibits periodic oscillation in its lowest energy state, meaning it moves and breaks time-translation symmetry without losing energy.
- Frohlich Life Story 1, Frohlich Life Story 2, Frohlich Hamiltonian: Herbert Fröhlich's pioneering theory that biological molecules, when driven by metabo energy, can condense into coherent macroscopic quantum states (akin to a roor temperature Bose-Einstein condensate).

Visionary Anirban

- Brain Jelly Q&A
- My TEDx talk: Triplet of triplet in the universe: In 2020 (and expanded in 2022) Bandyopadhyay's team demonstrated that brain-extracted microtubules exhibi self-similar resonance patterns across Hz, kHz, MHz, GHz, and THz frequenci effectively acting as polyatomic time crystals that project like holograms.
- Silicon AI has to go: "Suddenly, the dream of artificial brains is no longer scien fiction in the childish sense. It becomes an engineering horizon. The book poi toward a future in which light, sound, microwave signals, magnetic vortices, helical symmetries, and low-energy resonant materials interact as parts of a ne cognitive hardware. Not a giant server farm. Not a mechanical imitation of the cortex. But a different class of intelligence altogether — one that is distributed adaptive, co-evolving, and startlingly efficient."

Origins of Life

Hameroff has teamed up with Dante Lauretta, the Principal Investigator of NASA's OSIRIS-REx mission (which returned samples from the asteroid Bennu in 2023). Th working hypothesis? Polyaromatic hydrocarbons (PAHs) found in Bennu and the Murchison meteorite may exhibit time-crystal-like quantum oscillations. Bandyopadhyay's group has already simulated a Murchison organic molecule showi potential time-crystal behavior. If confirmed—and if these oscillations are dampen by anesthetics—it suggests that primitive, proto-conscious Orch OR-mediated qua may have been the organizing feedback that actually drove abiogenesis. Consciousr may have been the spark at the origin of life, not a late-stage emergent property.

Penrose Objective Reduction

The Gran Sasso constraint. Donadi, Piscicchia, Curceanu, Diósi, and Bassi ran a dedicated underground experiment at Italy's Gran Sasso lab searching for the faint ray emission that stochastic DP collapse should produce from germanium nuclei. T found nothing, tightening the DP cutoff parameter R_0 by three orders of magnitude enough to "rule out the natural parameter-free version of the Diósi–Penrose model" (*Nature Physics* 17:74, 2021). Figurato et al. (2024) tightened it further. Derakhshani, Diósi, Laubenstein, Piscicchia, and Curceanu applied the result to Orch OR in "At crossroad of the search for spontaneous radiation and the Orch OR consciousness theory" (*Physics of Life Reviews*, 2022), arguing the DP-stochastic variant is ruled out Kelvin McQueen's reply conceded the narrow refutation but noted the refuted varia "was never advocated by anyone" — Penrose's OR is non-stochastic, without the white-noise field the bound assumes. A December 2025 arXiv paper formalizes a nc local gravitational self-energy variant designed to evade the constraint. Stochastic collapse models are under pressure; core Penrose OR lives in narrower parameter space but isn't refuted.

The Fuentes program. Ivette Fuentes has become Penrose's main experimental collaborator. Her approach uses Bose–Einstein condensates to probe whether gravi

plays an active role in collapse. Howl, Penrose, and Fuentes (2019) argues "gravitizi
QT" is testable near the Planck *mass* scale — much closer to experimental reach tha
the Planck *length*. A single BEC in a superposition of two locations would decay on
timescale predicted by the OR formula, giving a cleaner signature than entangleme
based tests. Scaling is the bottleneck: current BECs reach millions of atoms; the tes
needs tens of trillions. Fuentes works with experimentalists Chris Westbrook (Paris
and Philippe Bouyer (Amsterdam) on the hardware. A recent Penrose × Fuentes conversation discusses the Folman "T-cubed" experiment and whether the equivale
principle holds in quantum mechanics.

The September 2025 consortium white paper. Most significant recent developmen
"A Spin-Based Pathway to Testing the Quantum Nature of Gravity" — a 55-author
white paper submitted to the European Strategy for Particle Physics and co-author
by Sougato Bose, Anupam Mazumdar, Penrose, Fuentes, Ron Folman, Gerard Milbu
Adrian Kent, Chiara Marletto, Vlatko Vedral, Jonathan Oppenheim, Hendrik Ulbri
and dozens of others. It proposes using nanodiamonds with NV centers and Stern-Gerlach forces to create macroscopic spatial superpositions and test whether gravit
can entangle two such crystals. Penrose as co-author signals OR is now embedded i
mainstream quantum-gravity experimental planning rather than sidelined.

Bottom line. The stochastic-noise variant of DP is dead. Penrose OR (we can drop t
Diosi now) is unaffected. BEC and nanodiamond experiments aim to test OR direct
within the decade; the technology is still a gap away, but mainstream physics now takes the question seriously enough to build dedicated consortia around it. Any positive OR detection would confirm the collapse mechanism Penrose proposes —
Hameroff claim that *biology uses it* remains separately testable via the anesthesia an
superradiance lines.

The Gödelian Argument

Penrose continues to lean in on his argument in the LLM age. In late 2025, he gave
keynote presentation at the Breakthrough Discuss conference explicitly titled *Why*

Intelligence Is Not a Computational Process. Penrose has also been explicitly linking tl non-computable understanding to the peculiar nature of time in quantum reality. H argues that a classical object's properties can be entirely ascertained, but a quantun object's reality behaves unusually with regard to time—retroactively.

Peter Koellner's 2018 *Journal of Philosophy* papers (I, II) set modern terms with the D system — a type-free truth + absolute provability framework — and concluded Penı fails because two key steps apply rules to indeterminate formulas. Antonella Corrac and Sergio Galvan responded inside Koellner's own framework (2022, 2024): restrict to arithmetic formulas, the disputed steps become legitimate. The partial result — any formalism captures all true arithmetic propositions known to humans, at least true non-arithmetic proposition still escapes it — is the most serious live defense o Penrose-shaped claim in contemporary logic. On the skeptical side, Stanisław Krajewski's "On the Anti-Mechanist Arguments Based on Gödel's Theorem" (*Studi Semiotyczne* 34:1, 2020) is the strongest recent negative case, but concedes two Gödelian corollaries: we cannot prove our own consistency, and we cannot fully describe our notion of natural number. That entire 2020 issue (with Avron, Cheng, Quinon) is the best survey of the state of play.

The LLM boom has raised the stakes. Landgrebe and Smith's *Why Machines Will Ne Rule the World* (Routledge 2022, second edition 2025) argues AGI is mathematically impossible via a complex-systems route compatible with Penrose without dependin on the Gödelian line. Within Orch OR, Wiest (2025) and Sergi et al. (2025) keep non-algorithmic understanding central — Sergi et al. framing it as a *logical requirement* motivating the search for a non-computational physical mechanism. Skeptics count that progress in mathematical AI (DeepMind's AlphaGeometry, AlphaProof's IMO 2024 silver) points toward systems that will pass Peter Millican's "Tutoring Test" fo Gödelian metamathematics; Selmer Bringsjord continues updating his "Refutation Penrose's Gödelian Case Against AI".

Worth flagging for readers who dismiss the argument on familiar grounds: most cri (Putnam, Feferman, LaForte, Minsky, Russell & Norvig, Churchland & Grush) targe the ambitious "humans are not Turing machines" thesis. Penrose himself retreated "Beyond the Doubting of a Shadow" to the weaker "human mathematicians are not using a *knowably sound* algorithm." This modest version — compatible with humans being Turing machines whose consistency we cannot establish from within — is exactly what Corradini-Galvan partially formalize and what Krajewski's concession leave standing. It's the version to defend. And Kerskens & Pérez (2022) — MRI sign consistent with proton-spin entanglement correlating with working memory performance — is a rarely-noted empirical wrinkle that shifts the prior on whether non-computational substrate is *available* to the cognition Gödelian arguments concern.

The formal-logic consensus still runs against Penrose (consensus doesn't equate to truth). But it has been forced to articulate itself with unusual precision, carving out defensible partial version of his claim exactly as computationalism's stakes have ris

Other Notes

- **Photosynthetic quantum coherence (updated 2024–2026).** The old "photosynthesis proves long-lived functional electronic coherence in warm biology" slogan went through some ups and downs: the 2017–2020 reinterpretation period substantially weakened that reading by shifting many long-lived oscillations from electronic to vibrational/vibronic origins. But the story did not collapse. Zhu et al. (Nature Communications 2024) reported room temperature exciton-vibrational coherence of about 500 fs in the allophycocyan trimer and proposed a protection mechanism based on quantum phase synchronization of collective vibrational modes, in which fast-dissipating antisymmetric modes are lost while correlated symmetric modes persist. Follov up work and a 2026 JPCB review strengthen a narrower claim: protein scaffold can help select and protect specific vibronic coherences across multiple photosynthetic systems. The contrast with phycocyanin 620, which has similar pigment-pair geometry yet shows incoherent transfer, argues that local scaffold/conformation matters. This does not restore the original long-lived electronic-coherence thesis, but it does strengthen the case for biologically structured, room-temperature exciton-vibrational coherence as a serious warm quantum-biology mechanism.

- **Engineered superradiance in viral capsids** (**Dragnea group, Indiana Universit** Brome mosaic virus shells decorated with Oregon Green fluorophores exhibit collective Dicke-type emission at room temperature, with coherent oscillations transient absorption persisting for hundreds of picoseconds. The effect require the ordered icosahedral template — disordering the scaffold eliminates it — supporting the broader claim that geometric regularity in protein assemblies ca sustain collective quantum-optical phenomena under warm, wet conditions. Caveats worth flagging: the chromophores are synthetic dyes attached external not endogenous aromatics; the ~100 ps coherence is still 4–5 orders of magnitu shy of Orch OR's millisecond requirements; and Dicke superradiance is a differ physical process than the gravitationally-induced state-vector reduction Penro Hameroff invokes. Still, the work belongs in the post-Tegmark landscape of evidence that naive room-temperature decoherence estimates systematically underestimate what structured biological scaffolds can support.
    - Tsvetkova et al., *Radiation Brightening from Virus-like Particles*, ACS Nano 13, 11401 (2019). https://pubs.acs.org/doi/10.1021/acsnano.9b04786
    - Holmes et al., *Ultrafast Collective Excited-State Dynamics of a Virus-Supported Fluorophore Antenna*, J. Phys. Chem. Lett. 13, 3237 (2022). https://pubs.acs.org/doi/10.1021/acs.jpclett.2c00262 (preprint: https://arxiv.org/abs/2202.01733)
- Some syntheses and proposed experiments by Justin Echternach: Dimensional Convergence at the Nanoscale: A Novel Synthesis of Experimental Evidence Justifying Direct Measurement of Quantum Coherence in Microtubules, Geometric Foundations for Collective Quantum Phenomena in

[Microtubules: Convergent Evidence from Nanophotonic Analogs].

- McQueen et al. (2026) prove that IIT-based consciousness-collapse models requ combinatorial operator proliferation, ruling out the 'simple and testable' frami Adlam et al. (2025) prove that purely-quantum agency violates no-cloning plus linearity. Neither directly refutes Orch OR — gravitational collapse doesn't tra experiential structure, and Orch OR is a quantum-classical hybrid — but both should sharpen how Orch OR partisans state the architecture.

Discussion about this post

Comments Restacks

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